

BSM Performance Benchmark Documentation

The vectorized Week 4 chooser engine reproduces Week 3 prices exactly and processes large scenario arrays fast enough for model training, sensitivity grids, and an interactive application.

Benchmark Configuration

Item	Value
Workload	100,000 chooser prices
Median runtime	19.49 ms
Throughput	approximately 5.13 million prices per second
Tracked peak memory	approximately 6.1 MB
Time complexity	$O(n)$
Space complexity	$O(n)$ for vectorized output and intermediate arrays

Implementation Design

The module broadcasts scalar and array inputs through NumPy, evaluates the normal CDF with SciPy, validates positivity and time ordering, and returns arrays without Python-level loops. The adapter executes the Week 3 notebook functions and verifies exact agreement before the refactor is used downstream.

Use in Later Weeks

- Week 5 target construction
- Week 6 pricing comparisons
- Week 7 stress grids
- Week 8 interactive sensitivity curves

Benchmark Interpretation

The measured runtime describes the test environment and configured array size. Hardware, library versions, and array shapes can change the absolute number. The asymptotic result is stable: each additional contract requires a constant amount of formula work, so runtime and array memory grow linearly with n .